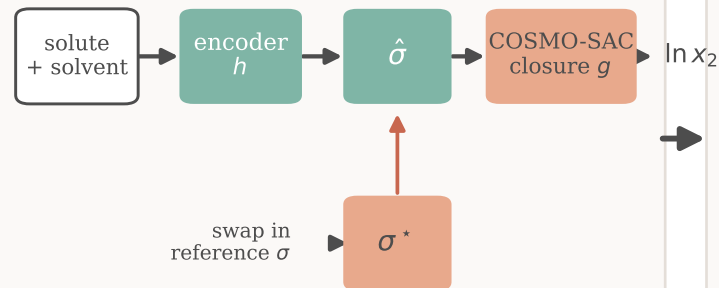


When do reference physical inputs help a learned solubility model?

1. The paradox

fixed, no fitted parameters



Premise: truer physical inputs $\rightarrow$ better.

Here, the opposite. So which is wrong — the closure, or the inputs?

2. The measurement

the reference input has an external oracle (VT-2005),

so the reference-input error splits exactly:

$$B = B_{\text{closure}} + B_{\text{insuff}}$$



$$B_{\text{closure}} > B_{\text{insuff}} \quad (P_{\text{boot}} \approx 0.78)$$

The ceiling is the CLOSURE, not the inputs.

3. No lever found

The obvious fix---a better closure---does not hold up on representative data.



No closure-fidelity lever established.

What the paper establishes, graded by evidence

CERTIFIED

- Paradox: reference σ makes it worse (3 seeds, $n = 5608$)
- Exact split $B = B_{\text{closure}} + B_{\text{insuff}}$ (Lemma 2)
- Our 2002 & 2010/dsp layers reproduce NIST ($0.003/10^{-4}$)

LIKELY

- Closure binds on low- γ set: $P_{\text{boot}} \approx 0.78$
- Attribution on the matched $n = 60$ corner (no transfer)
- Latent = low-rank, transferable surrogate ($n = 44$)

OPEN

- No fidelity lever: 2002 $\rightarrow$ 2010/dsp +3% ($n = 477$, $P = 0.62$)
- Transfer to the finite-composition regime
- pKa second closure as a fidelity cross-check